Animal Feed Science and Technology 228 (2017) 59–65

Contents lists available at ScienceDirect

Animal Feed Science and Technology

journal homepage: www.elsevier.com/locate/anifeedsci

Production performance, immunity, and heat stress resistance in Jersey cattle fed a concentrate fermented with probiotics in the presence of a Chinese herbal combination

Xiang Wang, Haijun Xie, Fu Liu, Yuhong Wang*

Biology Institute of Shanxi, No. 50, Shifan Street, Xiaodian District, Taiyuan City, Shanxi Province, 030006, China

ARTICLE INFO

Keywords:
Composite probiotics
Chinese herbal combination
Fermented concentrate
Jersey cattle

ABSTRACT

The goals of this study were to investigate the production performance, immunity, and heat stress resistance of Jersey cattle fed a concentrate fermented with compound probiotics in the presence of a Chinese herbal combination (CHC). First, the appropriate concentrations for the six Chinese herbs required to stimulate probiotic proliferation were investigated *in vitro*. Next, the optimal time and moisture content were determined for fermentation. We then tested the feeding effect of our fermented product on 36 growing cattle and 36 milking cows. The animals were randomly assigned to receive the product or not, resulting in half of each group receiving the treatment or serving as the control group. For 60 days, the experimental group was fed with the fermented product and the control group was fed with a basic concentrate that did not contain the probiotics or the Chinese herbs. The results showed that the cattle that received the fermented concentrate showed improved weight gain and milking cows that received the fermented concentrate showed increased average milk yield, milk fat content, and solid-not-fat content ($P < 0.05$). In addition, the serum levels of growth hormone, the 70 kDa heat shock protein (HPS70), and immunoglobulin G (IgG) of the cattle in the experimental group were significantly higher than the levels of cattle in the control group. A similar increase in the serum levels of HPS70, IgG, immunoglobulin A, and interferon γ was observed in milking cows that received the treatment ($P < 0.05$). No significant differences ($P > 0.05$) were observed in the serum levels of interleukin-2. Moreover, milking cows that received the fermented concentrate showed increased apparent digestibility of crude protein, ether extract, and crude fiber ($P < 0.05$). For the growing cattle, only crude protein digestibility was significantly increased in the experimental group ($P < 0.05$). Finally, we found that the ammonia content in the barn was decreased after the animals received the fermented concentrate. These results demonstrated that our fermented concentrate improved production performance, immunity, and heat stress resistance in Jersey cattle. We also demonstrated that this CHC and a mixture of probiotic strains acted as an ideal leavening agent for cattle's concentrate.

Abbreviations: ADF, acid detergent fiber; AIA, acid insoluble ash; BW, body weight; CF, crude fiber; CHs, Chinese herbs; CHC, Chinese herbal combination; CP, crude protein; DM, dry matter; EE, ether extract; GH, growth hormone; HSP70, heat shock 70 kDa protein; IgA, immunoglobulin A; IgG, immunoglobulin G; IFN-γ, interferon γ; IL-2, interleukin-2; NDF, neutral detergent fiber; SNF, solid not fat

* Corresponding author.
E-mail address: slw0728@126.com (Y. Wang).

http://dx.doi.org/10.1016/j.anifeedsci.2017.03.015
Received 26 July 2016; Received in revised form 22 March 2017; Accepted 24 March 2017
0377-8401/

1. Introduction

Antibiotics have played an important role in livestock production in the past few decades (Myaing et al., 2002). However, excessive use of antibiotics can lead to antibiotic resistance of pathogenic bacteria or result in antibiotic residue in animals and resulting animal products (Peng and Tan, 2015). Therefore, there is a significant need to exploit effective and safe substitutes for antibiotics in the feed industry. Microbial agent can be ideal antibiotic alternatives because it does not produce any residues and microbial resistance and pollutants to the environment (Wang, 2009). Animals can obtain both nutrients and the probiotics from microbial fermented feed (Hu, 2015). The probiotics in microbial fermented feed can prevent harmful microbe invasion in the digestive tract and enhance function of the immune system (Fu et al., 2014; Li et al., 2012). Moreover, fermented feed may provide more rich nutrition, better palatability, and increased digestibility (Niba et al., 2009). Due to these many advantages, feed that has been microbially fermented has drawn the attention of the feed industry.

Chinese herbs (CHs) such as astragalus, angelica, cowherb seed, codonopsis, licorice and rhizoma ligustici wallichii have been shown to enhance animal production performance or animal immunity as substitutes for antibiotics (Zhai et al., 2013). Previous studies have also demonstrated that these CHs are effective alone and have synergistic effects when combined (Wang, 2011). Thus, a combination of CHs may show improved efficacy when applied as a nutritional supplement in ruminant animal production. In addition, some studies have also shown that CHs can promote the proliferation of probiotics; these probiotics can also increase the release of CH active ingredients during the fermentation process (Liu et al., 2014). Therefore, use of these CHs together with probiotics as a leavening agent for feed would have broad potential application in the feed industry.

Immunity is the ability of animals to defend against pathogens and toxins and to avoid infections and diseases. Immunoglobulin G (IgG) and immunoglobulin A (IgA) are two major mammalian serum immunoglobulins (Clarke et al., 1985). IgG and IgA mediate humoral immunity and anti-infection immunity. Interferon γ (IFN-γ) and interleukin-2 (IL-2) are important cytokines and mediate cell immune response (Paust et al., 2011). High environmental temperatures cause heat stress that can negatively impact the health and production performance of cattle. Heat stress resistance is a self-protection reaction against high environmental temperatures and is positively correlated with the expression levels of the 70 kDa heat shock protein (HSP70) (Chenxi et al., 2013).

Applications of microbial-fermented concentrates have been widely studied in pigs and poultry. However, there have been limited studies of the application of microbial-fermented concentrates in ruminant animals. In addition, there has been few reports of using a Chinese herbal combination (CHC) together with probiotics in a fermented concentrate. Therefore, the goal of this study was to prepare an effective fermented concentrate and test it in Jersey cattle. Finally, our study provides an experimental basis for industrial application of a supplement of CH and probiotic strains in cattle.

2. Materials and methods

2.1. Strains and CHC

The compound probiotic liquid contained 1.5 mg/kg of *Bacillus subtilis* (our lab breeding strains), 0.5 mg/kg of *yeast*, and 1.5 mg/kg of *Lactobacillus acidophilus.* The liquid medium used for growing the bacteria mixture was composed of 10 g/L peptone, 5 g/L yeast extract, 20 g/L glucose and 2 g/L potassium hydrogen phosphate. The CHC contained six CHs: astragalus, angelica, cowherb seed, codonopsis, licorice, and rhizoma ligustici wallichii.

2.2. Effect of CHs on probiotic proliferation in vitro

First, the probiotic strains were grown in 250 ml of liquid medium at 37 °C for 12 h. Then, 1 ml of the compound probiotic liquid was added to 100 ml liquid medium containing different concentrations of CH (100, 50, 25, 10, 5, 2.5, or 1.25 g/L). Liquid medium without any CHs was used for the control. The mixtures were incubated at 37 °C with 150 ∼ 160 r/min shaking for 48 h. After 48 h, the cultures were diluted 10-fold and the concentration was determined by measurement of the optical density (OD600) relative to that of the culture medium with no bacteria or CHs. Each condition was tested in triplicate. The effect of CHs on probiotic proliferation was assessed based on the S/N value. The S/N value was calculated with the following formula: S/N = (OD600 of sample − OD600 of bank control)/(OD600 of negative control − OD600 of bank control), where S refers to sample, and N refers to negative control.

2.3. Microbial fermentation experiments

Fermentation experiments were performed using CHs, 2 ml compound probiotic liquid extract, 30 g brown sugar, 0.5 ug vitamin B1, 0.1 ug vitamin B2, and water for each kg of concentrate. First, the influence of moisture content on the protein content of the product was investigated. To do this, the moisture contents for the fermented concentrate were set at 300, 350, 400, 450, and 500 g water per kg concentrate. Then the concentrates prepared with different water conentrations were fermented for 72 h at 37 °C. Second, the influence of fermentation time on the protein content of the product was investigated. During fermentation, we collected the first sample at 48 h, and then every 24 h to 144 h. The quality of the fermented concentrates was assessed based on the crude protein (CP) content and crude fiber (CF) content.

Table 1
The composition and nutrient levels of basic concentrate.

Composition (g/kg DM)	Concentrate for growing cattle	Concentrate for milking Cows
Corn grain	540.0	540.0
Soybean meal	210.0	190.0
Wheat bran	200.0	220.0
Salt	10.0	10.0
Supplement	10.0[a]	10.0[b]
Limestone	15.0	13.0
Calcium hydrogen phosphate	–	2.0
Sodium bicarbonate	15.0	15.0
Nutrient levels (g/kg DM)		
NE (MJ/kg)	7.8	7.2
CP	157.0	152.0
NDF	121.0	127.0
ADF	270.0	266.0
Ca	6.5	7.8
P	4.6	5.4

NE: Calculated value [based on Ministry of Agriculture of P.R. China (2012)].

[a] The vitamin-mineral premix supplied per kg of concentrate: 0.072 g vitamin A (600,000 IU/g); 0.026 g vitamin D3 (200,000 IU/g); 2 g vitamin E (500IU/g); 1 g iron sulfate; 1.5 g copper sulfate; 1.25 g manganese sulfate; 4 g zinc sulfate; 0.05 g potassium iodide; 0.03 g sodium selenite; 0.01 g cobalt chloride; Rumen bypass amino acids, rumen bypass fat, urease inhibitors, antioxidants and composite carrier.

[b] The vitamin-mineral premix supplied per kg of concentrate: 0.072 g vitamin A (900,000 IU/g); 0.026 g vitamin D3 (300,000 IU/g); 2 g vitamin E (300IU/g); 4.8 g iron sulfate; 1.5 g copper sulfate; 3.9 g manganese sulfate; 4.4 g zinc sulfate; 0.042 g potassium iodide; 0.05 g sodium selenite; 0.024 g cobalt chloride; Rumen bypass amino acids, rumen bypass fat, urease inhibitors, antioxidants and composite carrier.

2.4. The feeding experiment

Thirty-six bred Jersey cattle (10 months of age; body weight (BW) 215 ± 2.4 kg) were randomly divided into two groups. The control group was fed with a basic concentrate that was not fermented (3.5 kg dry matter (DM)/day). The experimental group was fed with the fermented concentrate (3.5 kg DM/day). The composition and nutrient levels of the concentrates are shown in Table 1. Similarly, thirty-six milking Jersey cows (BW 480 ± 25 kg; 100 ± 45 day in milk) were randomly divided into two groups. The experimental group was fed with the fermented concentrate (5 kg DM/day) and the control group was fed with the basic concentrate (5 kg DM/day). In addition, growing cattle were given 20 kg of corn silage per day and milking cows were given 32 kg of corn silage per day. Each group was separately placed in a barn. Each animal was fed separately and had free access to water. The dietary treatment was performed for 60 days and the use of antibiotics was completely banned during the experimental period. All experimental protocols complied with the National Institutes of Health Guidelines for the Care and Use of Experimental Animals.

2.5. Sample collection and evaluation

The body weight was measured for the growing cattle after dietary treatments of 0, 30, and 60 days. The milk yields of all Jersey milking cows were recorded weekly with a measuring bottle, fishbone-type milking machine (JF, China). Milk samples were obtained from the four consecutive milkings collected on day 0, day 30, and day 60. Milk samples were stored at 4 °C and the analysis of total milk solids content, fat content, protein content, and lactose content were performed within 24 h after sample collection. Blood samples were collected on day 60 and centrifuged at 3000 × g for 10 min. Serum samples were collected and stored at −20 °C for the detection of IFN-γ, IL-2, IgA, IgG, GH, and HSP70. During the last week of the experiment, fresh fecal samples (about 100 g each sample) were collected three times per day by rectal grab sampling (morning, noon, and night) for five consecutive days. Fresh fecal samples were added to 10 ml of 1.07 g/ml dilute sulphuric acid and stored at −20 °C. After all fecal samples were collected, fecal samples were thawed at room temperature and later dried at 65 °C for nutrient analysis. The ammonia content in the barn was regularly detected by an ammonia suction pump (Z-800XP, USA).

2.6. Chemical analyses

The analysis of DM, organic matter (OM), ether extract (EE), CP, and CF in the concentrates were performed according to the AOAC (1990), using method number 934.01 for DM, 930.05 for OM, 920.39 for EE, 981.10 for CP, and 978.10 for CF. Acid detergent fiber (ADF) and neutral detergent fiber (NDF) were determined based on the procedure of Goering and Van Soest (1970).

Milk sample analysis was performed with the assistance of the Cows DHI testing center of the Animal Husbandry Genetic Breeding Center of Shanxi. The protein, fat, lactose, and solid-not-fat (SNF) concentrations in milk were determined by intermediate infrared spectroscopy (MIRIS DMA; MIRIS; Sweden) following procedure AOAC (1990).

Acid-insoluble ash (AIA) was used as an indigestible marker in digestibility studies. AIA content was determined based on the 4 mol/L HCl insoluble ash method described by Van Keulen and Young (1977). Proximate analysis was applied to the feed and feces to calculate the DM, CP, and CF digestibilities using the methods mentioned above. The digestibility was then calculated using the following formula: Digestibility = [1 − {(Nf × Ad)/(Nd × Af)}]; where, Nf = Nutrient concentration in feces (% DM), Nd = Nutrient concentration in diet (% DM), Af = AIA concentration in feces (% DM), and Ad = AIA concentration in diet (% DM).

The detection of blood indexes was carried out using ELISA kits (Sigma Chemical Co.). IFN-γ, IL-2, IgA, IgG, GH, and HSP70-standard and serum sample were serially diluted 2-fold. Then, 50 ul diluted standard and 50 ul diluted serum were added to the plates and incubated for 1 h at 37 °C. After washing 3 times with washing buffer, anti-bovine primary antibodies conjugated with biotin (Sigma Chemical Co.) were added to each well and incubated for 1 h at 37 °C. Then, after 3 washes with washing buffer, anti-biotin antibodies conjugated with horseradish peroxidase (Sigma Chemical Co.) were added and incubated at 37 °C for 1 h. Finally, after washing 3 times with washing buffer, avidin-biotin-peroxidase complex was added and incubated at 37 °C for 10 min. Ophenylenediamine was used as the substrate and absorbance was measured at 450 nm with an ELISA reader (Bio-Tek Instruments Inc., USA).

2.7. Statistical analysis

Statistical analysis was performed using SPSS 17. 0. The statistical significance was determined by one-way ANOVA followed by Duncan test. P-values less than 0.05 were considered statistically significant. Additionally, there is no relationship at all between the growing cattle and milking cows and the analysis has to be performed separately on each of them.

3. Results

3.1. CHs stimulated probiotic proliferation in vitro

To investigate the effect of CHs on the probiotic mixture, we first tested the six CHs individually for the ability to stimulate probiotic proliferation. The proliferation data is shown in Table 2. Co-incubation of the probiotic mixture with each kind of CH led to a significant level of bacteria proliferation in a dose-related manner. Astragalus, cowherb seed, codonopsis, and ligustici wallichii showed their best stimulatory potential in a concentration range of 2.5–5 g/L. In addition, angelica showed optimal stimulatory potential in a concentration range of 2.5–10 g/L, and the most effective concentration of licorice was in the range of 1.25–5 g/L.

3.2. The effects of moisture content and fermentation time on the fermented concentrate

As shown in Table 3, CP content and CF content were measured after 48 h, 72 h, 96 h, 120 h, and 144 h of fermentation. A high level of CP content was observed after fermentation for 120 h. With the addition of 450 g water per kg concentrate, the CP in the fermented concentrate showed maximum yield. In contrast, as water content increased, CF gradually decreased. Interestingly, the fermented concentrate showed improved nutrition, good color, strong acidic smell, and good palatability. Based on these results, we selected 120 h fermentation time and 450 g water per kg concentrate moisture content for the fermentation of the concentrate.

3.3. Effects the fermented concentrate on production performance in Jersey cattle

As shown in Table 4, after the 60 days feeding test, the average weight gain of the growing cattle in the experimental group was significantly higher than that of growing cattle in the control group ($P < 0.05$). For milking cows, the average milk yield, milk fat, content and SNF content of the experimental group were significantly improved relative to the control group ($P < 0.05$). However, there were no significant differences in protein content or lactose content ($P > 0.05$). The change of average milk-yield from week 1 to week 8 is shown in Fig. 1.

Table 2
The sample/negative control (S/N) value for assessing the proliferation effect of the CHs on compound probiotics *in vitro*.[1]

CHs name	S/N value						
	Different concentrations of CH[2]						
	100 g/L	50 g/L	25 g/L	10 g/L	5 g/L	2.5 g/L	1.25 g/L
Astragalus	0.69 ± 0.04^{c}	1.07 ± 0.03^{c}	1.34 ± 0.08^{b}	1.79 ± 0.07^{ab}	2.33 ± 0.08^{a}	1.92 ± 0.06^{a}	1.52 ± 0.05^{b}
Angelica	0.50 ± 0.02^{c}	0.72 ± 0.01^{c}	0.98 ± 0.05^{c}	2.02 ± 0.09^{a}	1.93 ± 0.07^{a}	1.74 ± 0.07^{a}	1.55 ± 0.04^{b}
Cowherb seed	0.61 ± 0.03^{c}	0.89 ± 0.02^{c}	0.92 ± 0.04^{c}	1.42 ± 0.06^{b}	2.04 ± 0.09^{a}	1.70 ± 0.05^{a}	1.29 ± 0.02^{b}
Codonopsis	0.58 ± 0.01^{d}	0.84 ± 0.03^{c}	1.08 ± 0.07^{c}	1.37 ± 0.08^{b}	1.95 ± 0.06^{a}	2.01 ± 0.08^{a}	1.53 ± 0.02^{b}
Licorice	0.70 ± 0.02^{c}	0.96 ± 0.04^{c}	1.13 ± 0.06^{c}	1.43 ± 0.09^{b}	1.71 ± 0.07^{a}	1.94 ± 0.05^{a}	2.47 ± 0.08^{a}
Ligustici wallichii	0.59 ± 0.03^{d}	0.95 ± 0.02^{c}	0.98 ± 0.03^{c}	1.10 ± 0.05^{c}	2.30 ± 0.08^{a}	1.69 ± 0.04^{ab}	1.41 ± 0.03^{b}

[a,b,c,d]Means within a row without a common superscript differ, $P < 0.05$.
CH:Chinese herb; Chs: Chinese herbs.
[1] S/N = S/N = (OD600 of sample − OD600 of blank control)/(OD600 of negative control − OD600 of blank control). S refers to sample, and N refers to negative control.
[2] Each treatment is repeated in triplicate.

Table 3
The effects of moisture content and fermentation time on fermented concentrate.

Item	Amount of added water per kg concentrate[1]				
	300 g	350 g	400 g	450 g	500 g
Crude protein (g/kg)	145.0 ± 9.8^{c}	155.6 ± 7.6^{b}	157.8 ± 3.6^{b}	169.6 ± 6.5^{a}	162.3 ± 8.7^{ab}
Crude fiber (g/kg)	89.7 ± 3.2^{b}	83.2 ± 4.5^{b}	77.5 ± 6.6^{a}	73.2 ± 5.5^{a}	73.6 ± 2.3^{a}
Item	Fermentation time[1]				
	48 h	72 h	96 h	120 h	144 h
Crude protein (g/kg)	155.0 ± 5.6^{c}	163.6 ± 7.2^{c}	171.8 ± 6.9^{b}	179.6 ± 8.9^{a}	172.3 ± 8.2^{b}
Crude fiber (g/kg)	90.7 ± 8.9^{b}	81.2 ± 2.6^{b}	79.1 ± 6.3^{b}	63.2 ± 2.5^{a}	63.6 ± 1.5^{a}

^{a,b,c}Means within a row without a common superscript differ, $P < 0.05$.
[1] Each treatment is repeated in triplicate.

3.4. The effects of the fermented concentrate on immunity, heat stress resistance and nutrient digestibility in Jersey cattle

To evaluate the effects of the fermented concentrate on immune function and heat stress resistance in Jersey cattle, the serum levels of GH, HPS70, IgG, IgA, IFN-γ, and IL-2 were detected. The data is shown in Table 5. For the growing cattle, the serum levels of GH, HPS70, and IgG in the experimental group were significantly higher than those in the control group ($P < 0.05$), but there were no significant differences in the levels of IgA, IFN-γ, or IL-2 ($P > 0.05$). For milking cows, the serum levels of HPS70, IgG, IgA and IFN-γ in the experimental group were significantly higher than those in the control group ($P < 0.05$).

After the 60 days feeding test, the nutrient digestibility of fermented concentrate was measured and the results are presented in Table 5. For the growing cattle, the CP digestibility of the fermented concentrate was significantly higher than that of the basic concentrate ($P < 0.05$), however, there were no significant differences in OM, EE, and CF ($P > 0.05$). For the Jersey cows, the EE, CP, and CF digestibility of the fermented concentrate were significantly higher than that of the basic concentrate ($P < 0.05$).

3.5. The change in the ammonia content in the barn

To investigate the effect of the fermented concentrate on the shelter environment, the ammonia content in the barn was measured. The results are shown in Table 6. The ammonia content of the barn that housed the treatment group was significantly lower than that of the barn that housed the control group ($P < 0.05$). In addition, the offensive odor of feces was decreased in the treatment group when compared with the control group.

4. Discussion

Concentrate supplements are indispensable for a healthy cow diet, because these supplements can provide part of the energy, protein, calcium, phosphorus, and vitamins that are not provided by the main diet. The use of an effective and efficient concentrate is important for the growth performance and milk performance of cows. Probiotics and CHs can act as antibiotic alternatives to promote health. In this study, we fermented the concentrate with a mixture of probiotics and CHC. We predict this fermented concentrate has potential applications for the cattle industry.

CHs such as astragalus, codonopsis, cowherb seed, rhizoma ligustici wallichii and liquorice, and angelica have been demonstrated

Table 4
Effects of the fermented concentrate on production performance in Jersey cattles.[1]

Item[1]	The experimental group	The control group
Dry matter intake, kg /d	7.65 ± 0.12^{a}	7.23 ± 0.09^{a}
Initial weight, kg	216.3 ± 2.4^{a}	213.8 ± 2.2^{a}
Final weight, kg	249.1 ± 1.9^{a}	241.3 ± 1.7^{b}
Average daily gain, kg /d	0.55 ± 0.02^{a}	0.46 ± 0.01^{b}
Average milk-yield, kg/d	16.93 ± 0.18^{A}	14.85 ± 0.13^{B}
Milk composition (g/kg)		
Fat	37.10 ± 0.21^{A}	30.20 ± 0.85^{B}
Protein	44.60 ± 1.52^{a}	41.20 ± 1.76^{a}
Lactose	43.20 ± 0.98^{a}	42.10 ± 1.15^{a}
Solid Not Fat	125.30 ± 4.39^{a}	112.50 ± 5.27^{b}

The values with unlike superscripts differ at $P < 0.05$ (small letters) or $P < 0.01$ (capital letters).
[1] Each mean represents 3 pens with 18 cattles each.

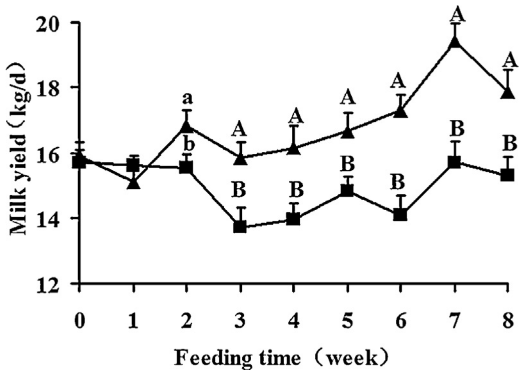

Fig. 1. The change of average milk-yield (kg/d) from week 1 to week 8. The experimental group was fed with the fermented concentrate (▲, n = 18). The control group was fed with the basic concentrate (■, n = 18). Each mean was marked with standard errors. Bars marked with different letters differ at $P < 0.05$ (small letters) or $P < 0.01$ (capital letters).

Table 5
Effects of the fermented concentrate on immunity, heat stress resistance and nutrient digestibility in Jersey cattle.[1]

Item[1]	Growing cattle		Milking cows	
	The experimental group	The control group	The experimental group	The control group
GH, ug/ml	6.96 ± 0.16^{a}	4.95 ± 0.03^{a}	5.12 ± 0.11^{a}	$4.89 \pm 0.0\ 9^{a}$
HPS-70, ng/ml	0.92 ± 0.02^{a}	0.86 ± 0.09^{b}	0.96 ± 0.03^{a}	0.79 ± 0.06^{b}
IgA, g/L	0.72 ± 0.04^{a}	0.59 ± 0.05^{a}	0.79 ± 0.06^{a}	0.62 ± 0.02^{b}
IgG, g/L	1.89 ± 0.07^{a}	1.49 ± 0.02^{a}	1.97 ± 0.05^{a}	1.88 ± 0.05^{b}
IFN-r, ug/L	2.46 ± 0.06^{a}	2.19 ± 0.08^{a}	2.56 ± 0.09^{a}	2.47 ± 0.05^{b}
IL-2, ug/L	1.56 ± 0.01^{a}	1.49 ± 0.08^{a}	1.57 ± 0.05^{a}	1.53 ± 0.03^{b}
Nutrient digestibility				
Organic matter	0.740 ± 0.014^{a}	0.653 ± 0.012^{a}	0.766 ± 0.016^{a}	0.667 ± 0.017^{a}
Crude protein	0.621 ± 0.011^{a}	0.533 ± 0.007^{b}	0.695 ± 0.007^{a}	0.597 ± 0.009^{b}
Ether extract	0.625 ± 0.010^{a}	0.546 ± 0.010^{a}	0.684 ± 0.015^{a}	0.572 ± 0.009^{b}
Crude fiber	0.595 ± 0.006^{a}	0.549 ± 0.008^{a}	0.650 ± 0.014^{a}	0.520 ± 0.007^{b}

[a,b]Means within a row without a common superscript differ, $P < 0.05$.
[1] Each mean represents 3 pens with 18 cattles each.

to promote animal health (Nie and Wang, 2010). Therefore, we used these six CHs as additives for the fermented concentrate. We first studied the proliferation effect of the CHs on compound probiotics in *vitro*. The proliferation of these probiotic strains increased significantly in a dose-dependent manner in response to these CHs. These results could provide a leading direction for fermentation experiment of concentrate.

Many studies have shown that production process conditions influence the production and quality of the fermented feed (Zhong et al., 2007), and determining the optimal conditions for fermentation is particularly important. Moisture in the environment affects the growth of microorganism, and in solid-state fermentation, the ratio of water to material must be appropriate (Sporleder et al., 2002). Too much water will make the material stick together, affecting permeability. Too little water limits the growth and metabolism of microorganisms and eventually leads to a reduction in nutrient production. The production of CP was highest at 450 g water per kg concentrate, indicating that this condition was best for microbial metabolism. We also found that the content of CP maintain did not change after 120 h, possibly due to nutrient limitation. Based on our results and consideration of actual production costs, the optimal fermentation time should be 120 h.

In animal production, the daily gain of growing cattle is a main index in the evaluation of performance. For milking cows, milk production and milk quality are the important evaluation indicators of dairy production performance (Bujko et al., 2011). In this study, after feeding with our fermented concentrate, the daily gain of growing cattle was significantly improved, and the average milk

Table 6
The ammonia content in the barn.

Item[1]	The experimental group	The control group
Ammonia content(mg/m^3)	0.72 ± 0.02^{b}	0.89 ± 0.03^{a}

[a,b]Means within a row without a common superscript differ, $P < 0.05$.
[1] Each mean represents 3 pens with 8 points.

yield of milk fat content and SNF content for milking cows was improved. These results may suggest that the concentrate fermented by the composite probiotics with this CHC improved growth performance and milking performance of Jersey cattle. This may be due to increased CP content in the fermented concentrate or may result from the production of bacterial proteins, organic acids, digestive enzymes, growth factors and other microbial products that may promote the digestion and absorption of nutrients. A third possible explanation for the observed improved performance may be that the probiotics of the fermented concentrate inhibit the activity of harmful microbes that can decrease animal production. In addition, the observed effects could result from the maximum effect of the CHs that are stimulated by the probiotics and improve production performance of Jersey cattle (Cao et al., 2004).

In this study, the levels of IgG, IgA, and IFN-γ in the cattle serum were significantly improved in the experimental group. These results demonstrated that our fermented concentrate improved cell-mediated immunity and systemic humoral immunity. Interestingly, we also found that the level of GH in the serum of growing cattle was significantly improved. These results indicate that feeding cattle the microbial-fermented concentrate has important health effects as evidenced by the changes in hormone levels. The increased level of HSP70 in the serum of the experiment group indicated that consuming the microbial-fermented concentrate may improve the heat stress resistance of Jersey cattle. The enhanced immunity and heat stress resistance in the experimental group are likely due to the CHC that enhance immunity or may be attributed to effects of the probiotics that can act as nonspecific immunity factors to stimulate and activate the host's immune cells.

Modern standardized farming seeks to develop a more ecological breeding practice and achieve low emissions (Yang, 2011). The feeding of the fermented concentrate can reduce ammonia emissions, possibly due to the effects of the probiotics that promoted nutrient digestion and absorption. In addition, the reduction of ammonia and other irritating substances in the intestinal contents decreased feces odor, improving the breeding environment. Thus, this study provides the theoretical and experimental basis for the development and application of a more ecological farming model.

5. Conclusions

In conclusion, we demonstrated that the use of a fermented concentrate containing CHs and probiotics improved cattle health. We determined the appropriate fermentation time and moisture content for preparation of this fermented concentrate. Finally, we demonstrated that use of this CHC together with this composite probiotic produced an effective fermented concentrate. We showed that use of this concentrate improved production performance, immunity, and heat stress resistance in Jersey cattle. Additionally, use of this fermented concentrate improved the barn environment and promoted overall health, similar to the use of antibiotics, to some degree. In brief, our study provides the experimental basis for industrial application of a fermented supplement for cattle.

Conflict of interest

The authors have no conflicts of interest to declare.

Acknowledgment

This work was supported by Shanxi province science and technology research of Youth Fund (201601D202063).

References

Association of Official Analytical Chemistry (AOAC), 1990. Official Methods of Analysis, 15th ed. AOAC International, Arlington.
Bujko, J., Kocman, R., Žitny, J., Trakovická, A., Cyril, H., 2011. Analyse of traits of milk production in dairy cows. Potravinarstvo 5, 5–9.
Cao, G.W., Zhou, S.L., Jiang, Y.K., 2004. Effect of the chinese traditional medicine's extractant on multiplication of lactobacillus. Sichuan Anim. Vet. Sci. 11, 30–31.
Chenxi, M.A., Wang, L., Zhu, B., Liu, C., Katsuzumi, O., 2013. Effects of heavy metal on expression of heat shock protein 70(hsp70) in hela cells. Acta Laser Biol. Sinica 80, 154–164.
Clarke, G.N., Stojanoff, A., Cauchi, M.N., Johnston, W.I.H., 1985. The immunoglobulin class of anti-spermatozoal antibodies in serum. Am. J. Reprod. Immunl. 7, 143–147.
Fu, R., Chen, R., Huang, Y., Zhao, H., Li, H., 2014. Effects of fermentated protein feed on the antioxidant ability and immune ability of growing-finishing pigs. Modern. J. Anim. Husbandry Vet. Med. 2, 21–24.
Goering, H.K., Van Soest, P.J., 1970. Forage fiber analyses (apparatus, reagents, prcedures, and some applications). USDA Agric. Handbook 379, 1–20.
Li, G., Brown, P.J., Tang, J.X., Xu, J., Quardokus, E.M., Fuqua, C., Brun, Y.V., 2012. Surface contact stimulates the just-in-time deployment of bacterial adhesins. Mol. Microbiol. 83, 41–51.
Hu, Xin-Xu, 2015. Effects of complex probiotics fermented feed without antibiotic on growth performance,plasma biochemical parameters,immune function and meat quality in growing-finishing pig. J. Huazhong Agric. Univer. 1, 72–77.
Liu, F., Han, C., Liu, C., Yanfei, H.E., Zhu, L., Chen, S., 2014. Progress of microbial fermentation chinese medicine feed additive. Amino Acids Biotic Res. 36, 18–22.
Myaing, T.T., Saleha, A.A., Arifah, A.K., Raha, A.R., 2002. Public health aspects of antibiotics use in farm animals. J. Agric. For. Livest. Fish. Sci. 8, 146–163.
Niba, A.T., Beal, J.D., Kudi, A.C., Brooks, P.H., 2009. Potential of bacterial fermentation as a biosafe method of improving feeds for pigs and poultry. Afr. J. Biotechnol. 8, 1758–1767.
Nie, J.P., Wang, X., 2010. Application of additives of CHM in commodity broiler young chicken. Anim. Husbandry Feed Sci. 31, 21–22.
Paust, H.J., Ostmann, A., Erhardt, A., Turner, J.E., Velden, J., Mittrücker, H.W., 2011. Regulatory t cells control the th1 immune response in murine crescentic glomerulonephritis. Kidney Int. 80, 154–164.
Peng, X., Tan, Z. 2015. Feed additive for animals of P-thymol, salt derivative or ester derivative thereof. US, US9018256.
Sporleder, R.A., Linden, J.C., Schroeder, H.A., Johnson, D., Henk, L.L., Tengerdy, R.P. 2002. Process for the production of nutritional products with microorganisms using sequential solid substrate and liquid fermentation. US, US 6444437 B1.
Van Keulen, J., Young, B.A., 1977. Evaluation of acid-insoluble ash as a natural marker in ruminant digestibility studies. J. Anim. Sci 44, 282–287.
Wang, J.H., 2009. Research status of micro-ecological feed additives and their application in livestock and poultry production. Anim. Husbandry Feed Sci. 30, 36–38.
Wang Ji. 2011. A galactagogue mdicine and its preparation method: CN, CN 101695524B [P].
Yang, H., 2011. Ecological breeding technique of rural chicken. Beijing Agric. 33, 72–75.
Zhai, J.F., Guo, D.X., Tian, H., 2013. Research and application of chinese herbal feed additives in animal production. Pratacultural Sci. 30, 1131–1134.
Zhong, C., Huang, H., Qin, X., 2007. Study on process conditions of feeding protein production by the fermentation of pineapple pericardium. Feed Ind. 11, 54–57.